

Biomedical Named Entity Recognition Using a Hybrid BiLSTM-Transformer-CRF Architecture

Final Project Report

CSCI-E-89b: Introduction to Natural Language Processing

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Abstract

Biomedical text mining requires structured extraction of disease names and related clinical concepts from raw text. Named Entity Recognition (NER) enables this transformation by assigning BIO labels to each token in a sequence. This project develops a modern, hybrid sequence labeling architecture that integrates a Bidirectional Long Short-Term Memory network (BiLSTM), a transformer encoder block, and a Conditional Random Field (CRF) decoding layer. The objective is to construct a workflow using the NCBI Disease Corpus, demonstrating how contemporary deep learning methods can be implemented for biomedical entity extraction.

The work proceeds through dataset preprocessing, tokenization, BIO span alignment, vocabulary construction, architecture definition, CRF-based structured decoding, and quantitative evaluation.

Two Jupyter notebooks accompany the report. The first provides end-to-end training and evaluation. The second offers an error analysis and visualization suite, including confusion matrices, CRF transition heat maps, and a four-quadrant diagnostic visualization of boundary errors, label errors, hallucinations, and correct predictions.

Generative AI Disclosure

Portions of this project were developed with reference to Generative AI tools (Copilot; ChatGPT, OpenAI, 2025 version), used in accordance with the course AI Technologies Policy including conceptual clarification, methodological guidance, validating code logic, identifying and resolving exceptions, and generating text and code segments. All substantive decisions regarding problem formulation, dataset selection, model design, feasibility analysis, and implementation strategy are my own. All AI-generated code was independently reviewed, debugged, and validated for correctness before inclusion.

1. Introduction

Extracting structured biomedical knowledge from scientific literature is essential for downstream applications in drug discovery, clinical decision support, and large-scale epidemiological analysis. Disease mentions in text often appear with varied morphology, complex boundary patterns, and contextual dependencies. Biomedical NER addresses these challenges by assigning BIO labels to each token in a sequence.

NER is framed as a conditional sequence labeling problem. Given input tokens $x_{1:T}$, the objective is to infer the most likely tag sequence $y_{1:T}$:

$$\hat{y}_{1:T} = \arg \max_{y_{1:T}} p(y_{1:T} | x_{1:T}).$$

This project combines a BiLSTM encoder with a transformer block to capture both local and long-range dependencies, followed by a CRF layer to enforce valid BIO transitions during decoding. This architecture aligns with modern NER research while remaining computationally efficient for CPU-only execution.

2. Dataset

The NCBI Disease Corpus provides annotated disease mentions with character-level spans across PubMed abstracts. Its structure requires careful preprocessing because biomedical terminology frequently includes hyphens, slashes, and multi-token constructs.

The notebooks implement tokenization, character-to-token span alignment, BIO conversion, and validation procedures to ensure that gold spans align precisely with token boundaries. These steps form the foundation of the sequence labeling pipeline.

Download the NCBI Disease Corpus from:

<https://huggingface.co/datasets/bigbio/bc5cdr/tree/main>

Unzip and extract requisite files from the following structure:

BC5CDR_Docs_Data\CDR_Data\CDR.Corpus.v010516\CDR_TestSet.PubTator.txt

BC5CDR_Docs_Data\CDR_Data\CDR.Corporus.v010516\CDR_TrainingSet.PubTator.txt

BC5CDR_Docs_Data\CDR_Data\CDR.Corporus.v010516\CDR_DevelopmentSet.PubTator.txt

3. Installation and Configuration

To reproduce this project, install Python 3.10 or 3.11 and the following dependencies:

```
pip install tensorflow==2.15
```

```
pip install tensorflow-addons
```

```
pip install numpy pandas matplotlib seaborn
```

```
pip install scikit-learn
```

```
pip install seqeval
```

```
pip install spacy
```

```
python -m spacy download en_core_web_sm
```

Training Dataset Download and CWD-Based Loading:

The model notebook load all dataset files from the current working directory (CWD). No path configuration is required. Users should place train.tsv, dev.tsv, and test.tsv in the same folder as the notebooks.

DATA_DIR = os.getcwd() ensures platform-independent reproducibility.

4. Methodology

The methodological framework for this project follows the structure of a modern sequence labeling system. It proceeds through preprocessing, architecture construction, model training under a structured conditional likelihood, and quantitative evaluation paired with qualitative diagnostics.

4.1 Preprocessing Pipeline

Accurate preprocessing is essential for biomedical NER, where disease names frequently exhibit complex multi-token structure, hyphenation, and nonstandard punctuation. The preprocessing logic performs the following steps:

Tokenization:

Text is segmented into word-level tokens using standard whitespace and punctuation boundaries. Biomedical text introduces nontrivial edge cases, particularly with multi-word noun phrases and parentheses, so downstream span alignment procedures validate each token boundary carefully.

Character-to-Token Span Alignment:

The NCBI Disease Corpus provides entity annotations as character offsets. To train a sequence model, each character span must be mapped precisely to the corresponding token indices. The alignment routine iterates through token boundaries, identifies tokens whose character ranges overlap known entity spans, and validates that concatenating token substrings reproduces exactly the gold annotation text.

Any mismatch (for example, due to punctuation or tokenization differences) is flagged to prevent silent training errors.

BIO Tag Construction:

Each token receives one of three labels:

- B-DISEASE — the first token of a disease mention.
- I-DISEASE — continuation tokens of the same entity.
- O — all tokens not part of a disease span.

BIO encoding introduces structural constraints that are later reinforced through the CRF decoding layer.

Vocabulary Construction:

Token sequences are converted into integer indices. A vocabulary is built directly from the training corpus without external embeddings to maintain reproducibility. Similarly, a mapping from BIO labels to integers (label2id and id2label) is stored for later decoding.

Sequence Padding and Masking:

Neural architectures require uniform sequence lengths within a batch, so the pipeline pads sequences to a fixed maximum length then applies masks so the CRF and transformer components ignore padded positions.

This ensures mathematically valid likelihood computation and avoids artificial transitions at padded boundaries.

Span Alignment Validation:

Before training, the notebook reconstructs each disease mention by concatenating the tokens selected by the alignment logic and checks for strict equality with the original gold entity. This step was critical during development, as biomedical corpora contain spans whose boundaries can be difficult to preserve after tokenization. This validation eliminated an entire class of downstream training inconsistencies.

4.2 Model Architecture

The model integrates recurrent, attention-based, and structured prediction components into a single sequence labeling system. This hybrid approach leverages the strengths of each paradigm.

Embedding Layer: Each token index is transformed into a dense vector representation. These embeddings are learned from scratch during training to reflect the specialized vocabulary of biomedical text.

Bidirectional LSTM Encoder: A BiLSTM processes the embedding sequence in both forward and backward directions such that:

$$\vec{h}_t = \text{LSTM}(x_t, \vec{h}_{t-1}), \vec{h}'_t = \text{LSTM}(x_t, \vec{h}'_{t+1}).$$

The contextual representation is the concatenation:

$$h_t = [\vec{h}_t; \vec{h}'_t].$$

In this way the models BiLSTM captures local sequential dependencies and short-range contextual structure characteristic of biomedical terminology (for example, multi-token disease names).

Transformer Encoder Block: The BiLSTM outputs are passed into a transformer encoder layer to capture longer-range dependencies. The self-attention mechanism allows each token to reference information from distant parts of the sentence, which is valuable in biomedical text where disease mentions often span multiple tokens or appear within complex constructions.

Linear Emission Layer: Each contextual vector h_t is mapped to a set of emission scores over the BIO labels. These scores represent the model's local preference for assigning each possible tag at position t .

CRF Decoding Layer: A Conditional Random Field (CRF) models the full tag sequence by learning transition preferences between labels. It combines the per-token emission scores with learned transition weights to compute a global sequence score.

During inference, the Viterbi algorithm selects the highest-scoring tag sequence, ensuring that predicted labels follow valid BIO patterns and improving boundary consistency.

4.3 Training Procedure

The training process jointly optimizes the BiLSTM encoder, transformer layer, and CRF decoder within a single sequence labeling framework. Model parameters are updated using the Adam optimizer, and all experiments run entirely on CPU for a modest number of epochs. Checkpoints record the embedding matrix, BiLSTM weights, transformer encoder weights, and the CRF transition matrix to ensure full reproducibility. Throughout training, sequence masks are applied so that padded tokens do not influence likelihood calculations or the CRF's transition structure.

4.4 Evaluation

Model performance is evaluated using both quantitative metrics and qualitative inspection. Quantitatively, the system is assessed by token-level accuracy and by entity-level precision, recall, and F1, computed through exact span matching.

Qualitatively, the companion visualization notebook extends the analysis with a confusion matrix of BIO tags, a visualization of the CRF transition matrix, and a span-level categorization of prediction errors. These span-level outcomes include correct predictions, boundary errors, label errors, and hallucinated entities. A four-quadrant radial diagram summarizes these error

categories into a single coherent representation. Together, these diagnostics reveal boundary inconsistencies and occasional hallucinations in dense abstracts - characteristic behaviors of biomedical NER models

5. Performance Overview

The hybrid BiLSTM-Transformer-CRF hybrid architecture produced stable performance across the development and test sets of the NCBI Disease Corpus. The Disease class achieved an F1 score of approximately 0.60, with precision around 0.69 and recall near 0.53. This distribution system identifies disease mentions conservatively, favoring precision over more expansive span predictions - expected behavior with CRF decoding that imposes strict boundary regularization on token-level outputs.

Recovery of Multi Token Disease Entities

The model performed reliably on multi token disease names, which dominate the corpus. Entities such as *tricuspid valve regurgitation*, *congestive heart failure*, and *congenital heart disease* were frequently reconstructed in full, including in longer clinical sentences - likely a result of model's contextual encoders (ie the BiLSTM captures immediate syntactic structure while the transformer layer provides additional context integration).

Boundary Errors and Structural Observations

Boundary inconsistencies constituted the largest class of errors. Span-level comparisons revealed occasional truncation or slight extension of disease mentions, particularly in repeated or syntactically dense contexts. Examples include partial predictions involving repeated occurrences of *tricuspid regurgitation* or incomplete coverage of *atrial flutter*. Importantly, the confusion matrix indicates that these errors arise primarily at the B-I boundary rather than from misclassification into the O label.

The CRF transition matrix further clarifies the model's internalized structure. High transition weights for $B \rightarrow I$ and $I \rightarrow I$ transitions demonstrate that the model learned coherent span continuation, while invalid transitions such as $I \rightarrow B$ were overtly suppressed. These learned patterns align with theoretical expectations of BIO-constrained decoding and directly support the behavior observed in predicted spans.

Aggregate Error Characterization

The visualization suite's radial four-quadrant summary consolidates prediction outcomes into correct spans, boundary errors, label errors, and hallucinations. Boundary errors dominate the distribution, consistent with the earlier quantitative discussion. Label errors and hallucinations are comparatively rare and tend to follow interpretable patterns, such as labeling *depression* as Disease in contexts where the gold annotation refers specifically to *neurologic depression*. These outcomes are probably more a reflection of domain-inherent ambiguity rather than instability in the model's learning dynamics.

6. What Worked and What Did Not

Effective Components of the Architecture

Several architectural decisions contributed measurably to overall performance. The BiLSTM encoder provided stable token-level contextualization that supported reliable recognition of short and medium-length disease mentions. The Transformer layer added the ability to incorporate longer-range dependencies, which improved coverage of multi token entities appearing in broader syntactic contexts. The CRF decoder played a central corrective role by enforcing valid BIO transitions and reducing fragmentation across predicted spans.

Limitations and Sources of Error

Boundary resolution remained the most persistent limitation. Although CRF decoding mitigates many issues, the system occasionally under-predicted the length of disease spans or merged repeated mentions into a single entity. These tendencies contributed directly to the lower recall observed in the quantitative results. Preprocessing also proved critical. The model's performance was highly sensitive to accurate character-to-token alignment, which required careful validation to prevent misaligned labels during training.

A final, less frequent class of errors involved hallucinated spans, typically arising from partial overlap between predicted and gold phrases. These errors underscore the challenge of modeling disease mentions without pretrained embeddings.

7. Lessons Learned

Preprocessing as a Determinant of Model Validity

Extensive preprocessing is essential for biomedical NER. Accurate span alignment is not a peripheral implementation detail but a determinant of model fidelity. Biomedical terminology often includes irregularities (ie punctuation, multi-word constructs) that require explicit handling to preserve the integrity of supervision signals.

Advantages of Hybrid Contextualization

The combination of a BiLSTM and a transformer encoder provided a balanced approach - the BiLSTM supported strong local coherence while the Transformer captured dependencies that extend beyond local neighborhoods. This hybrid structure functioned effectively on CPU and proved suitable where, in this case, computational efficiency is a constraint.

The Role of Visualization in Model Assessment

Metrics alone can obfuscate subtle forms of model behavior. Visualizations produced such as the confusion matrix, CRF transition heat map, span-level comparison tables, and the radial summary clarified the nature and distribution of errors. These visual diagnostics provided insight into both the stability of the model's predictions and the types of refinements that might yield the most benefit.

8. Conclusion

The hybrid BiLSTM-Transformer-CRF system demonstrated consistent and interpretable performance on the NCBI Disease Corpus. Its strengths lie in the accurate reconstruction of multi token disease mentions and in the structured regularization provided by CRF decoding. The model's primary limitation, namely boundary inconsistency, reflects a challenge inherent to biomedical NER tasks rather than instability in the architecture.

The results reinforce the applicability of hybrid sequence architectures in specialized domains. When pretrained biomedical transformers are unavailable or impractical, carefully configured BiLSTM-Transformer hybrids remain viable and computationally efficient.

Overall, the project illustrates a complete and reproducible pipeline for biomedical named entity recognition (NER) and provides a clear account of the architectural, preprocessing, and analytical considerations required to build such a system. The combination of quantitative results and qualitative diagnostics offers a comprehensive view of model behavior and suggests practical avenues for continued refinement.

Video Presentation:

The final project video presentation can be found [here](#) or via the URL:

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https://harvard.zoom.us/rec/play/vTXn_FgF7njoMpwqZkaC-PFPT-y5-6XCglUFkE8c5-  
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